



DEM 1110

Introduction to Demography

Data Presentation

PRESENTATION OF SOCIAL, ECONOMIC AND DEMOGRAPHIC DATA



COUPLES
FAMILY
SEXUALITY



BIRTH
FERTILITY
CONTRACEPTION



LIFE EXPECTANCY
MORTALITY
HEALTH



WORLD
MIGRATIONS
DISCRIMINATION
INTEGRATION



HOUSING
ENVIRONMENT
TERRITORY



GENDER
INEQUALITIES



THE STAGES OF
THE LIFE CYCLE
AGEING



Basic Measures

Basic Measures

- **RATIO**

- The Quantitative relation between two amounts showing the number of times one value contains or is contained within the other. Mathematically, a ratio shows the relative sizes of two or more values.

Examples of ratios include;

- Sex Ratio- The number of males per 100 females
 - $SR = (M/F) * 100$
- Dependency Ratio- The number of dependents per 100 economically active population



Basic Measures

- Dependency Ratio- The number of dependents per 100 economically active population

$$\text{DependencyRatio} = \frac{\text{Population}(0 - 14) + \text{Population}65 +}{\text{Population}15 - 65} \times 100$$



Basic Measures

- **Proportion**

- A proportion is a part or share of a whole; it is a ratio which expresses the relative size of a number in terms of total. It can also be defined as a part, share or number considered in comparative relation to a whole.

- Examples include;

- Femininity ratio, Masculinity ratio, Proportion of HIV+ persons eligible for ART initiation.

- **Proportion of women below 35 years**= Number of Women below age 35/Total number of women

Basic Measures

- **Percentage**

- A proportion expressed as a base of two or more numbers. The constant used is 100 for easy interpretation and comparison.

- **Examples include:**

- % of total population with access to education
 - % of total population with access to Health
 - % of total population with access to clean water



Basic Measures

- **Rate**

- A ratio of one figure to another in a specified period of time.

- **Examples Include;**

- $\text{Pregnancy Rate} = \frac{\text{Number of pregnancies}}{\text{Total Population of women(15-49)}} * 1000$
 - Sometime a rate is referred to as an occurrence or exposure ratio because the numerator is a number of occurrences or event usually in a year while the denominator measures the number of people exposed to the risk of experiencing the event.



Methods of Presenting Data

- Recap on types of Data!!!!
 - **Continuous Data:** These are numerical data that are measured in an unbroken scale such as age, weight and CD4 count. Often these data can be categorized during data analysis.
 - Age: . . .18, 19, 20, 21, 22, 23 . . .
 - Weight: . . .150, 151, 152, 153 . . .
 - Temperature: ... 37, 37.2, 37.9, 38...



Methods of Presenting Data

- Recap on types of Data!!!!
 - **Categorical:** These are data that can be broken into distinct categories, such as gender and marital status.
 - Age Categories: 18-24, 25-44, 45-64, 65+
 - Sex: Male, Female
 - HIV status: positive, negative



When is it useful to visually display data?

- Communicate with policy makers
- Document disease
- Demonstrate successes
- Show trade-offs between 2 choices
- Make decisions
- Other examples?

What do you want to communicate?

- Disease Trends?
- Difference among populations?



How do we convince people to listen to us?

SHOW THEM THE DATA!

Measles data in a table

Disease	District	Treatment	Date Seen	Date of onset	Sex	Result
Measles	Lusaka	Kanya	4-Jan-10	4-Jan-10	M	neg
Measles	Lusaka	Kanya	5-Jan-10	5-Jan-10	M	neg
Measles	Lusaka	Kanya	6-Jan-10	6-Jan-10	M	neg
Measles	Lusaka	Kanya	7-Jan-10	7-Jan-10	F	neg
Measles	Lusaka	Chawa	8-Jan-10	8-Jan-10	F	neg
Measles	Lusaka	Kanya	11-Jan-10	11-Jan-10	M	neg
Measles	Lusaka	Kanya	12-Jan-10	12-Jan-10	F	neg
Measles	Lusaka	Kanya	12-Jan-10	12-Jan-10	F	neg
Measles	Lusaka	Georg	14-Jan-10	14-Jan-10	F	neg
Measles	Lusaka	Georg	14-Jan-10	14-Jan-10	F	neg
Measles	Lusaka	Chawa	14-Jan-10	14-Jan-10	F	neg
Measles	Lusaka	Kanya	15-Jan-10	15-Jan-10	M	neg
Measles	Lusaka	Kanya	17-Jan-10	17-Jan-10	F	neg
Measles	Lusaka	Kanya	17-Jan-10	17-Jan-10	F	neg
Measles	Lusaka	UTH	27-Jan-10	26-Jan-10	M	neg
Measles	Lusaka	UTH	28-Jan-10	28-Jan-10	M	neg
Measles	Lusaka	Kamwa	28-Jan-10	28-Jan-10	M	Positive
Measles	Lusaka	UTH	29-Jan-10	29-Jan-10	F	neg
Measles	Lusaka	UTH	1-Feb-10	1-Feb-10	M	neg
Measles	Lusaka	Kanya	3-Feb-10	3-Feb-10	F	neg
Measles	Lusaka	Kanya	3-Feb-10	3-Feb-10	M	neg
Measles	Lusaka	Kanya	4-Feb-10	4-Feb-10	M	neg
Measles	Lusaka	UTH	5-Feb-10	5-Feb-10	F	Positive

Improving Presentation of Measles Data

<i>Disease</i>	<i>District</i>	<i>Treatin g HC</i>	<i>Date Seen</i>	<i>Date of onset</i>	<i>Sex</i>	<i>Result</i>
Measles	Lusaka	Kanya	4-Jan-10	4-Jan-10	M	neg
Measles	Lusaka	Kanya	5-Jan-10	5-Jan-10	M	neg
Measles	Lusaka	Kanya	6-Jan-10	6-Jan-10	M	neg
Measles	Lusaka	Kanya	7-Jan-10	7-Jan-10	F	neg
Measles	Lusaka	Chawa	8-Jan-10	8-Jan-10	F	neg
Measles	Lusaka	Kanya	11-Jan-10	11-Jan-10	M	neg
Measles	Lusaka	Kanya	12-Jan-10	12-Jan-10	F	neg
Measles	Lusaka	Kanya	12-Jan-10	12-Jan-10	F	neg

What is the main point of these data?

How can you make the data easier to interpret?

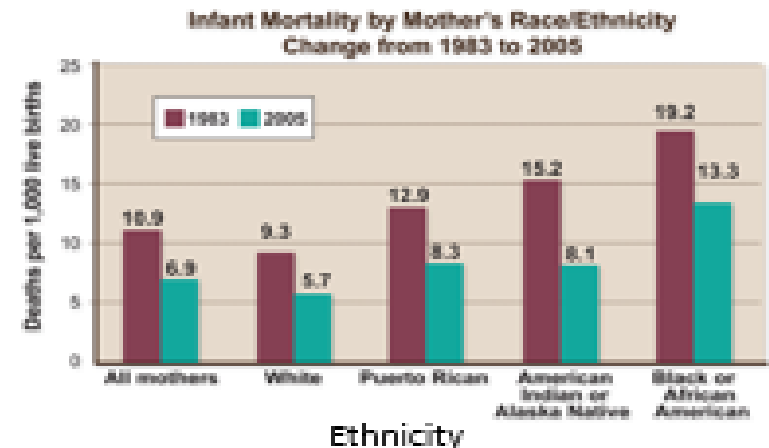
How could we present these data to show context?

Common ways to display data – tables and charts

Tables

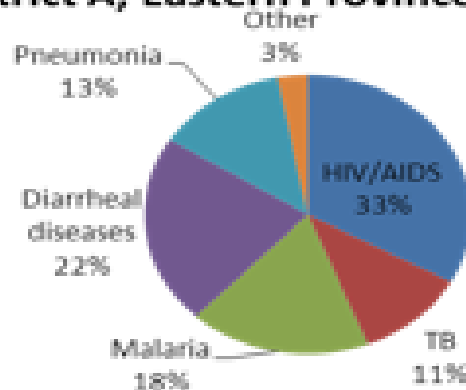
Program	Number of New In-Service Positions
Master of Medicine	9
Master of Nursing	10
BSc Nursing	345
Environmental Health	20
Midwifery	50
Theater Nursing	10
Clinical Officer – Anesthesia	10
Master of Public Health	15

Bar charts



Pie Charts

Distribution of Notifiable Diseases in District A, Eastern Province, July 2011



Line charts

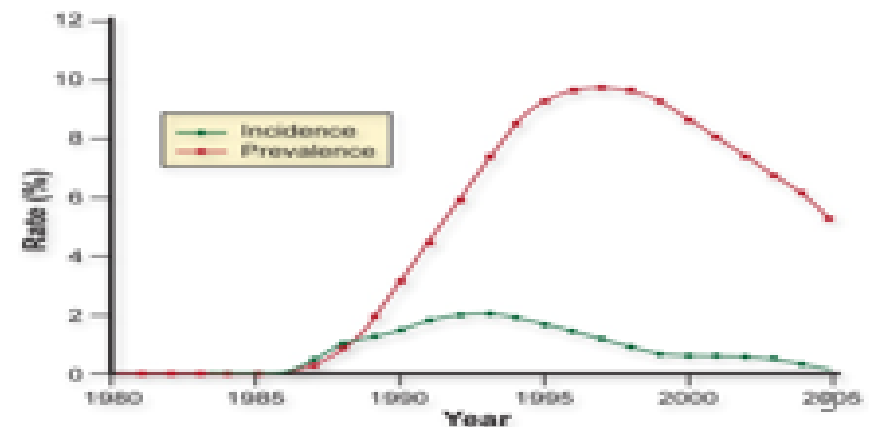


Table: An arrangement of data in rows and columns

Table 1.0: # PLWHA by District and Sex, Province Y ← Title

District	Male	Female	Total
A	14,800	200	15,000
B	400,000	20,000	420,000
C	997,000	3,000	1,000,000
D	985,000	15,000	1,000,000
E	1,460,000	40,000	1,500,000
F	465,000	35,000	500,000
G	940,000	10,000	950,000
H	380,000	220,000	600,000
I	900,000	600,000	1,500,000
J	545,000	5,000	550,000
Total	7,086,800	948,200	8,035,000

← Row

Persons Living With HIV/AIDS (PLWHA)

↑
Column



Improving Presentation of our Data

DOH ID#	School	Age	Sex	complaint	Have meds	Date visit ('04)	Temp	previous asthma visits
1	A	8	F	asthma	Y	3-Dec	98	1
2	B	11	M	asthma	N	8-Dec	102	0
3	B	10	F	asthma	Y	15-Dec	95	3
4	A	8	F	asthma	N	16-Dec	100	0
5	A	9	F	asthma	Y	18-Dec	101	1
6	B	11	M	asthma	Y	2-Dec	99	3
7	C	12	F	asthma	Y	12-Dec	103	2
8	A	10	M	asthma	Y	8-Dec	98	0
9	B	12	F	asthma	N	5-Dec	100	0
10	C	11	M	asthma	N	19-Dec	98	0
11	A	8	F	asthma	Y	20-Dec	98	1
12	A	9	M	asthma	N	12-Dec	102	2
13	B	10	F	asthma	Y	20-Dec	99	1
14	C	11	M	asthma	N	17-Dec	100	1
15	D	12	F	asthma	Y	4-Dec	102	0
16	F	9	F	asthma	N	2-Dec	98	0
17	D	9	F	asthma	N	7-Dec	101	0
18	C	7	F	asthma	N	14-Dec	97	1

- What is the main point of these data?
- How can you make it easier to read?
- Is a table the best way to represent these data?

CHARTS

- When used appropriately, can summarize and display complex data clearly and effectively and emphasize specific points
- Let you identify and present distributions, trends, and relationships among the data
- Help make sense of the data in the profile and communicate findings to planning groups and decision makers
- However, poorly designed or executed tables and figures can mislead users or distract them from your message!



Key points to consider when using charts to display data:

- What information is being conveyed in each?
- What different points do they make?
- Is this the most appropriate chart type represent these data?



Key elements of a Chart

Title

Vertical Axis (y-axis)



Legend

Horizontal Axis (x-axis)



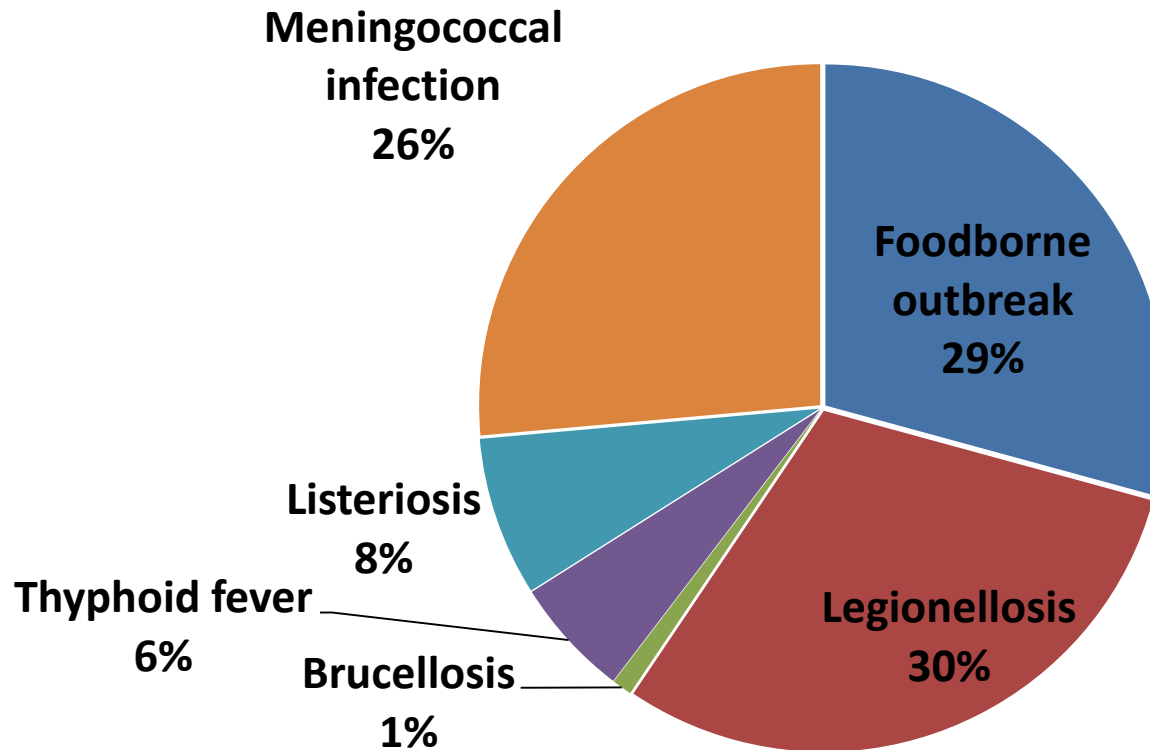
Pie Charts

- Circular chart split into segments which show components of a larger group
- Suitable for displaying categorical data, or discrete data in distinct categories
- The size of a “slice” is proportional to the amount of data (e.g., number of cases) it represents
- Proportion (%) of the total each component represents is frequently added on the slice
- Different colors can be used to identify various slices



Pie Charts

**Distribution of notifiable diseases reported
during football World Cup, June 4 - July 10, 1998**



Data source: Norwegian Institute of Public Health, Oslo, Norway



When **not** to use a pie chart

- When you would have more than 8 slices
- When the values of each slice are similar because it is difficult to see differences between slice sizes
- When you have a limited “n” or total (i.e., only two cases of measles out of 12 people)
- To compare with another pie chart

Bar charts

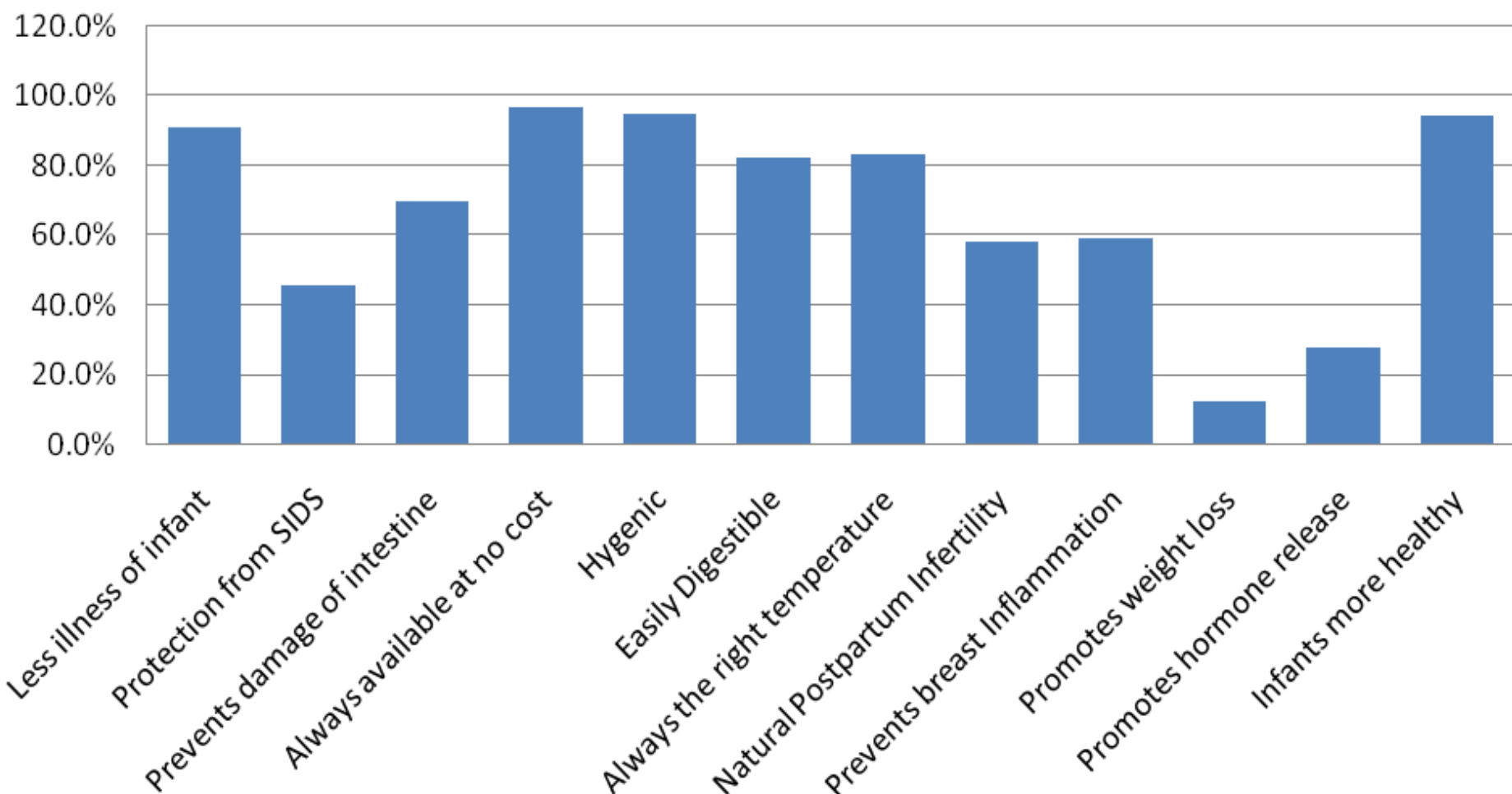
- Suitable for displaying categorical data and to compare discrete data in distinct categories
- Each bar represents one category
- Can be organized horizontally or vertically
- Height of the bars are proportional to the number of events (e.g. cases) in the category
- Variables in a bar graph can be discrete (e.g. sex, region, race) or continuous (e.g. age) but organized in categories (e.g. age groups)





Bar charts

Benefits of Breastfeeding



Limitations of Bar Charts

- Poorly designed chart can mislead users or distract them from your message!
- Can be difficult to see actual numbers

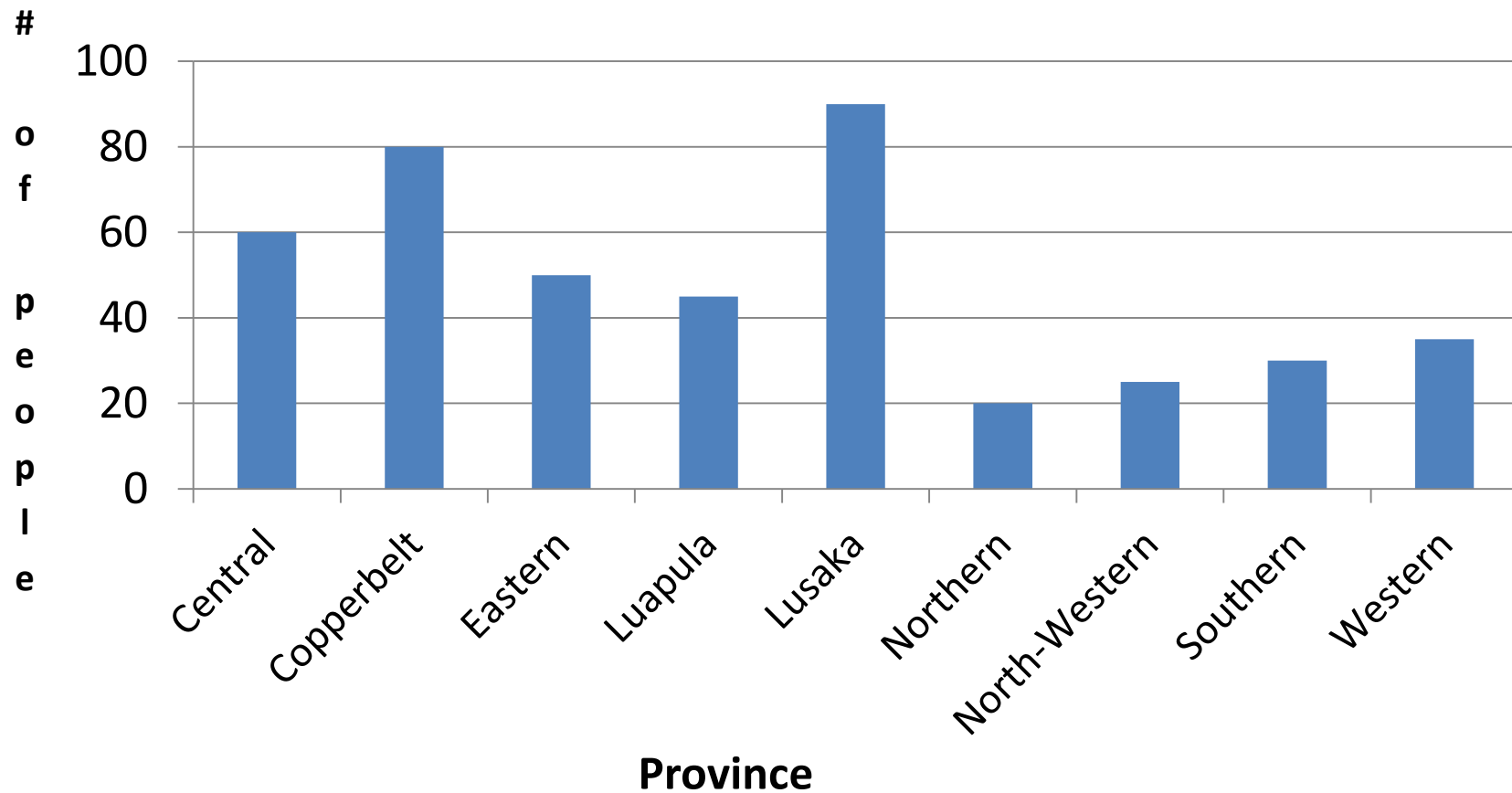
Four common types of bar charts:

- 1) Simple
- 2) Grouped
- 3) Stacked
- 4) 100% component



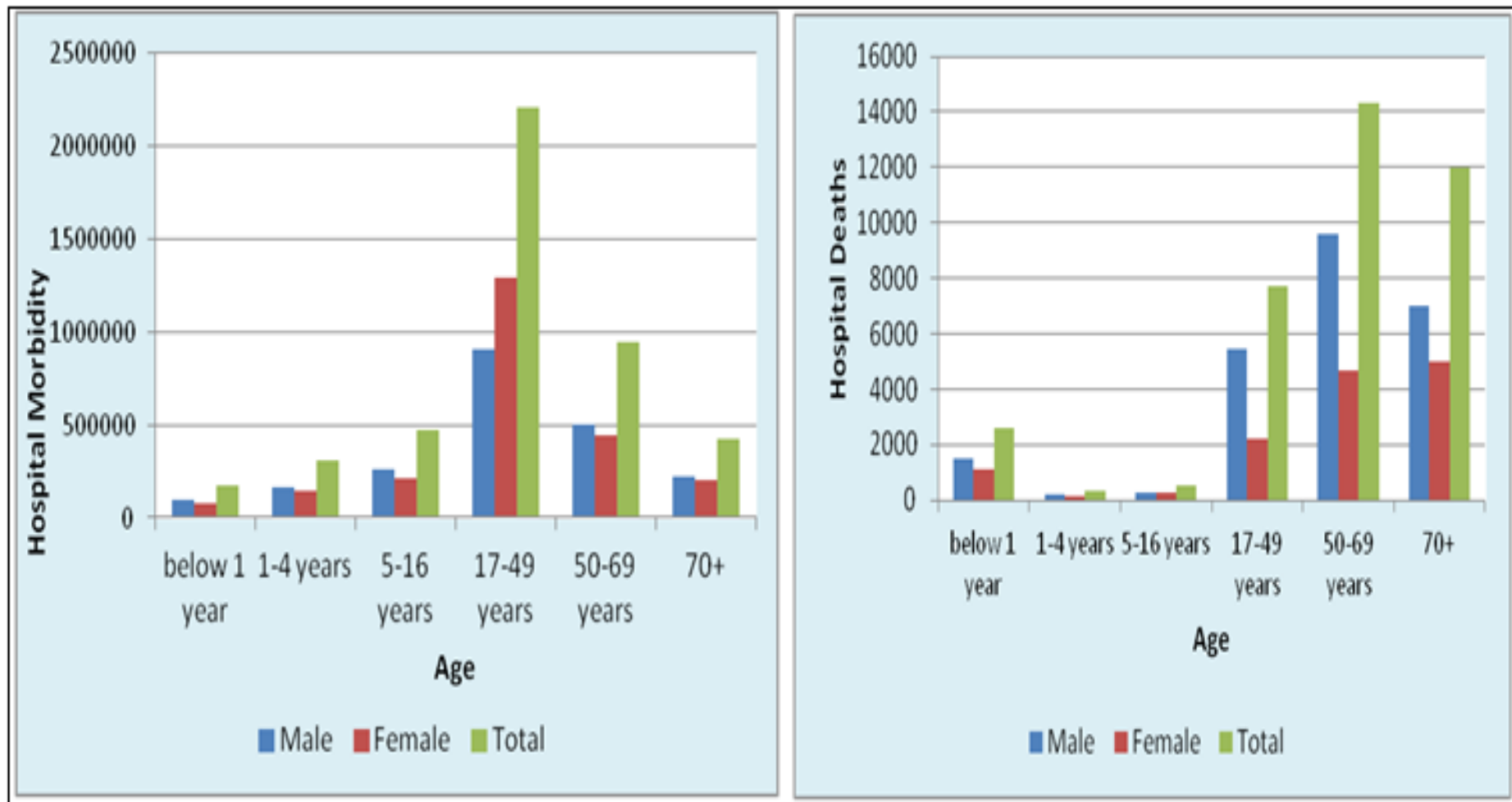
Simple bar chart

Number of district-level workers trained in EDU by province
in Zambia, 2012

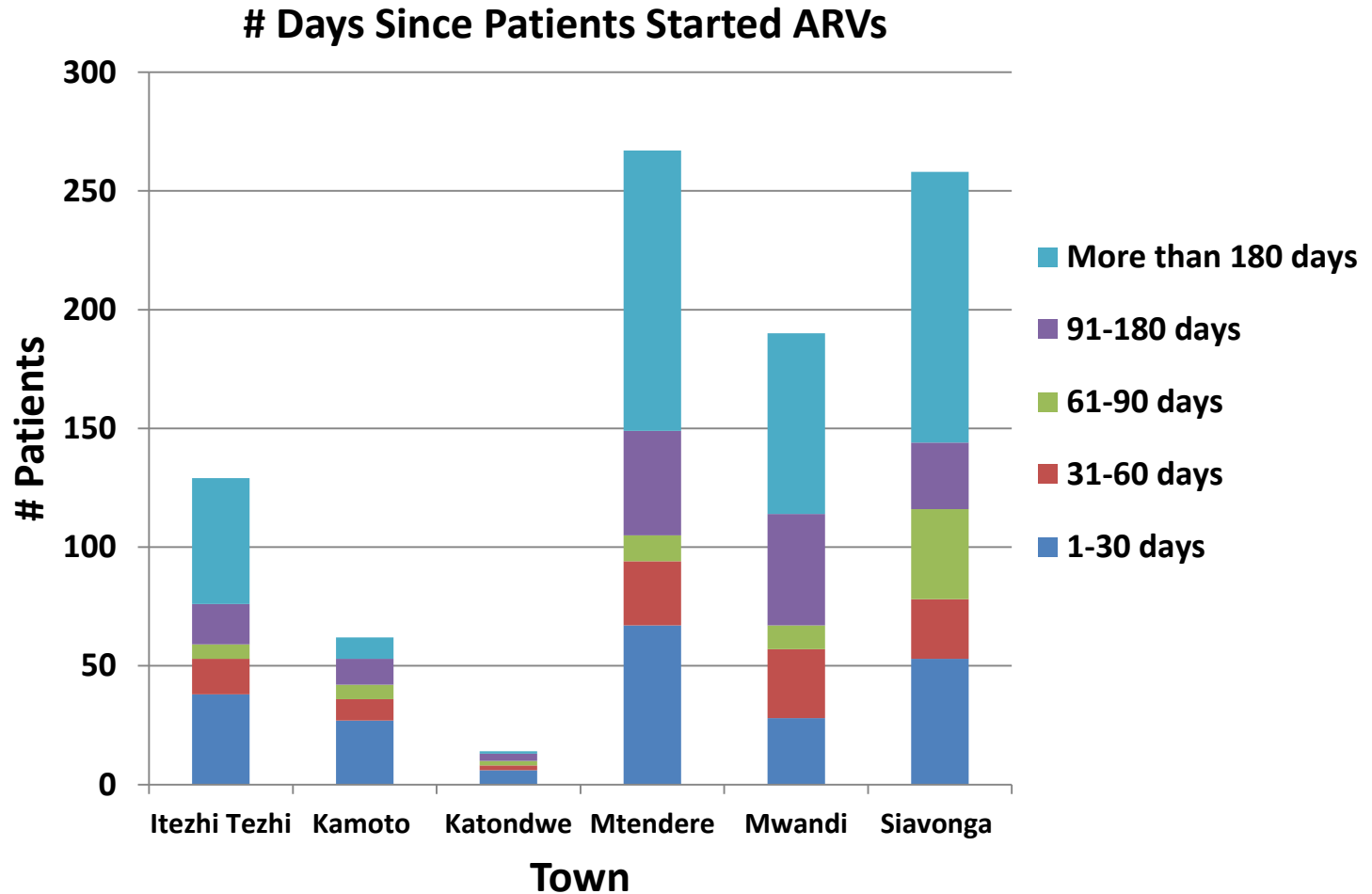


“Grouped” bar graph

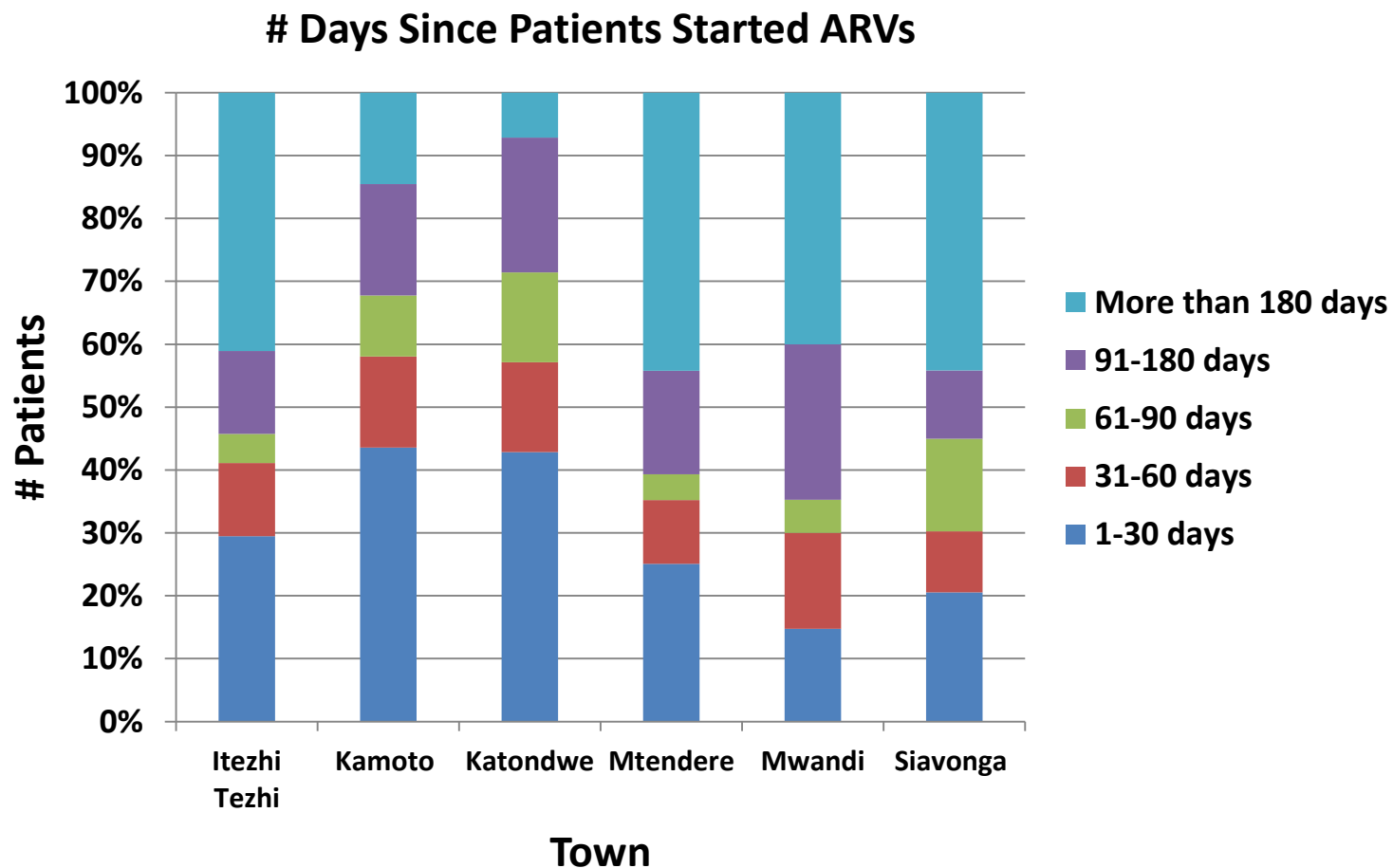
Figure: Distribution of Hospital morbidity and Hospital deaths in year by Age



Stacked bar graph



100% Component column graph



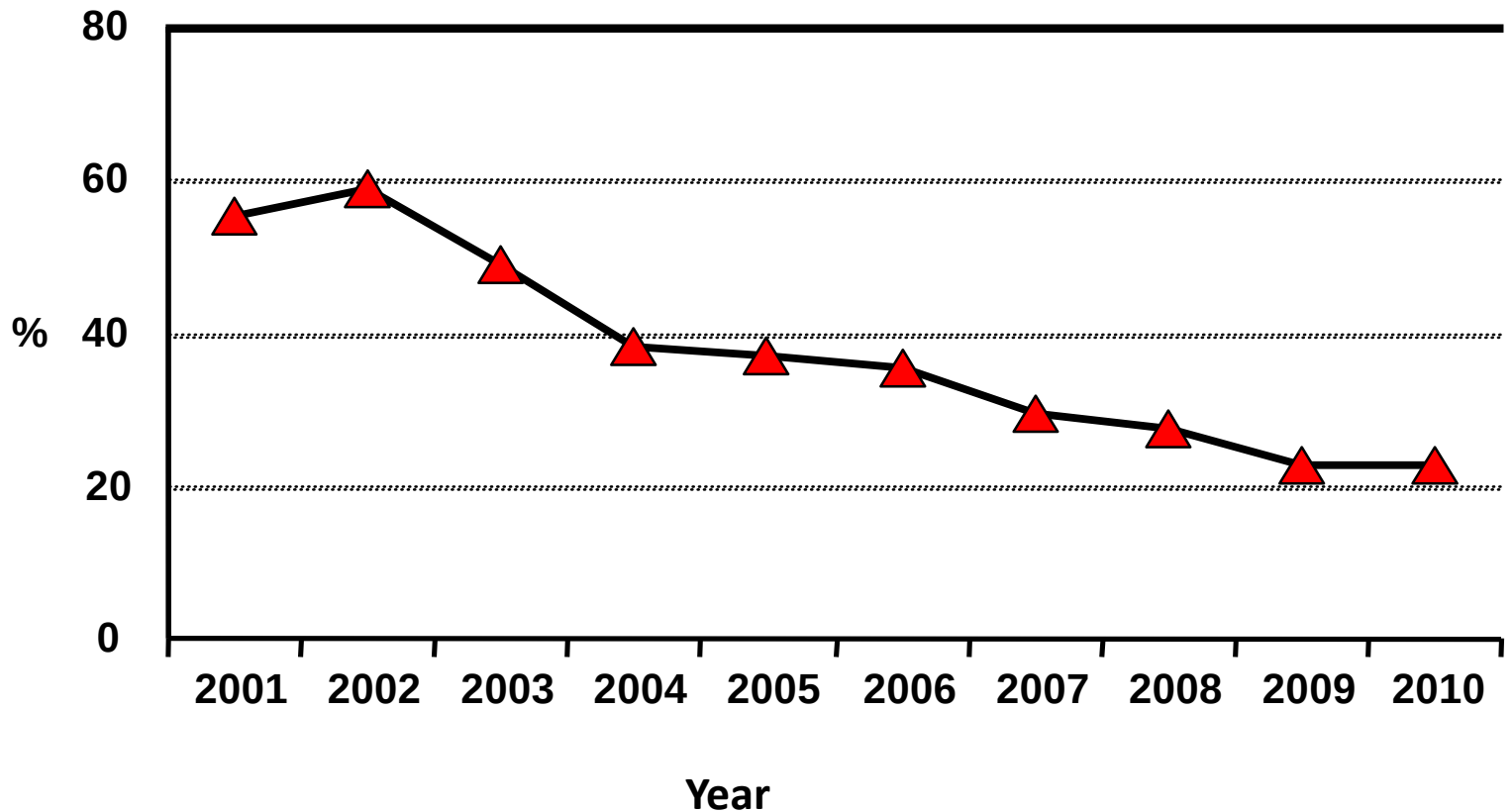
Line Graphs

- **Used for continuous variables**
- **Display relationships between 2 variables**
- **When used appropriately, can summarize and display complex data clearly and effectively and emphasize specific points**
- **Let you identify and present distributions, trends, and relationships among the data**



Line Graphs

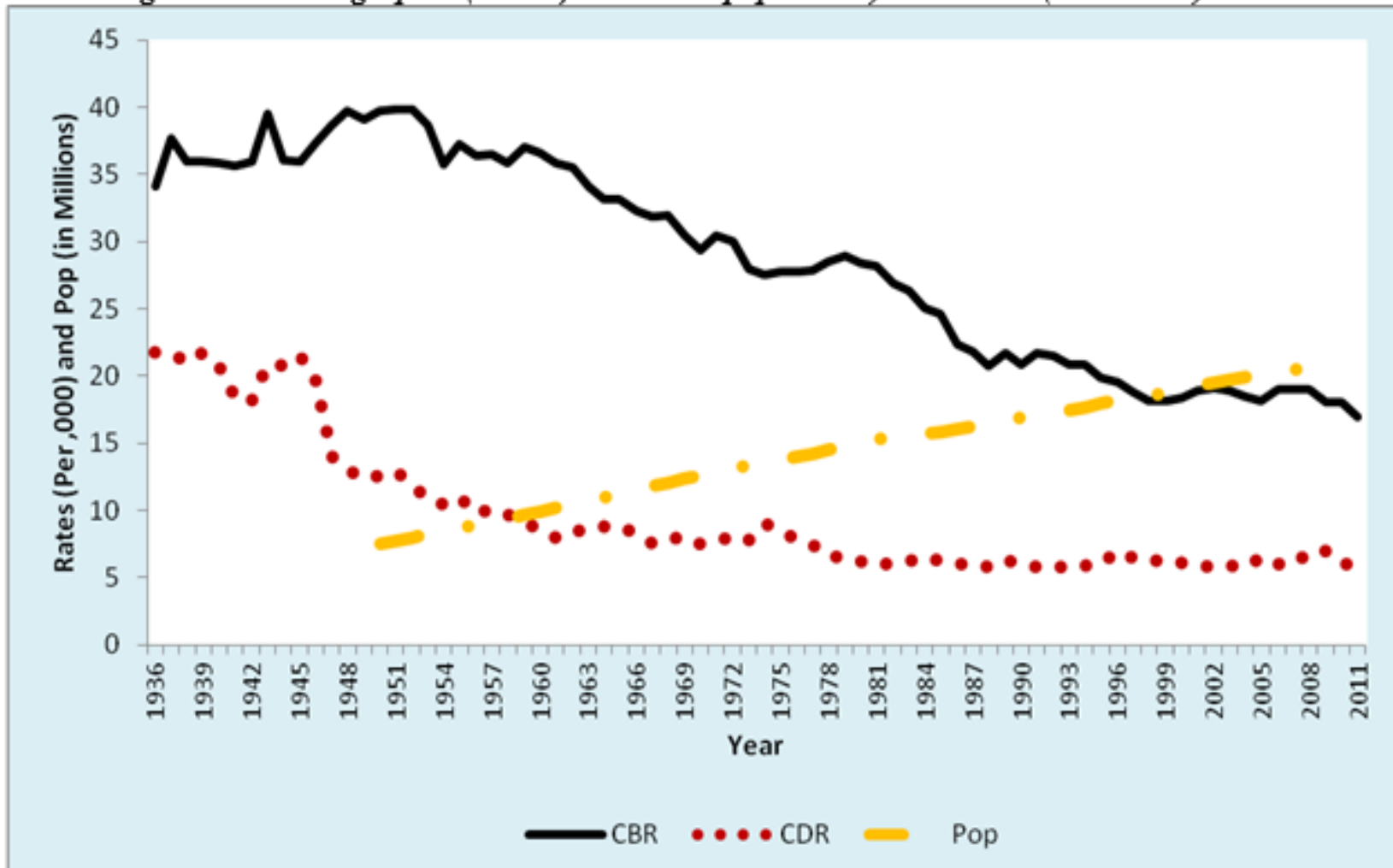
Trends in HIV prevalence among 15-49 year olds in province X: 2001– 2010



source: Ministry of Health annual report 2010

Line Graphs

Figure 2.1: Demographic (Births, Death and population) Transition (1936-2011)



Source of Data: US Census Bureau Website (<http://www.census.gov/population/international/data/idb/estandproj.php>)

Limitations of Line Graphs

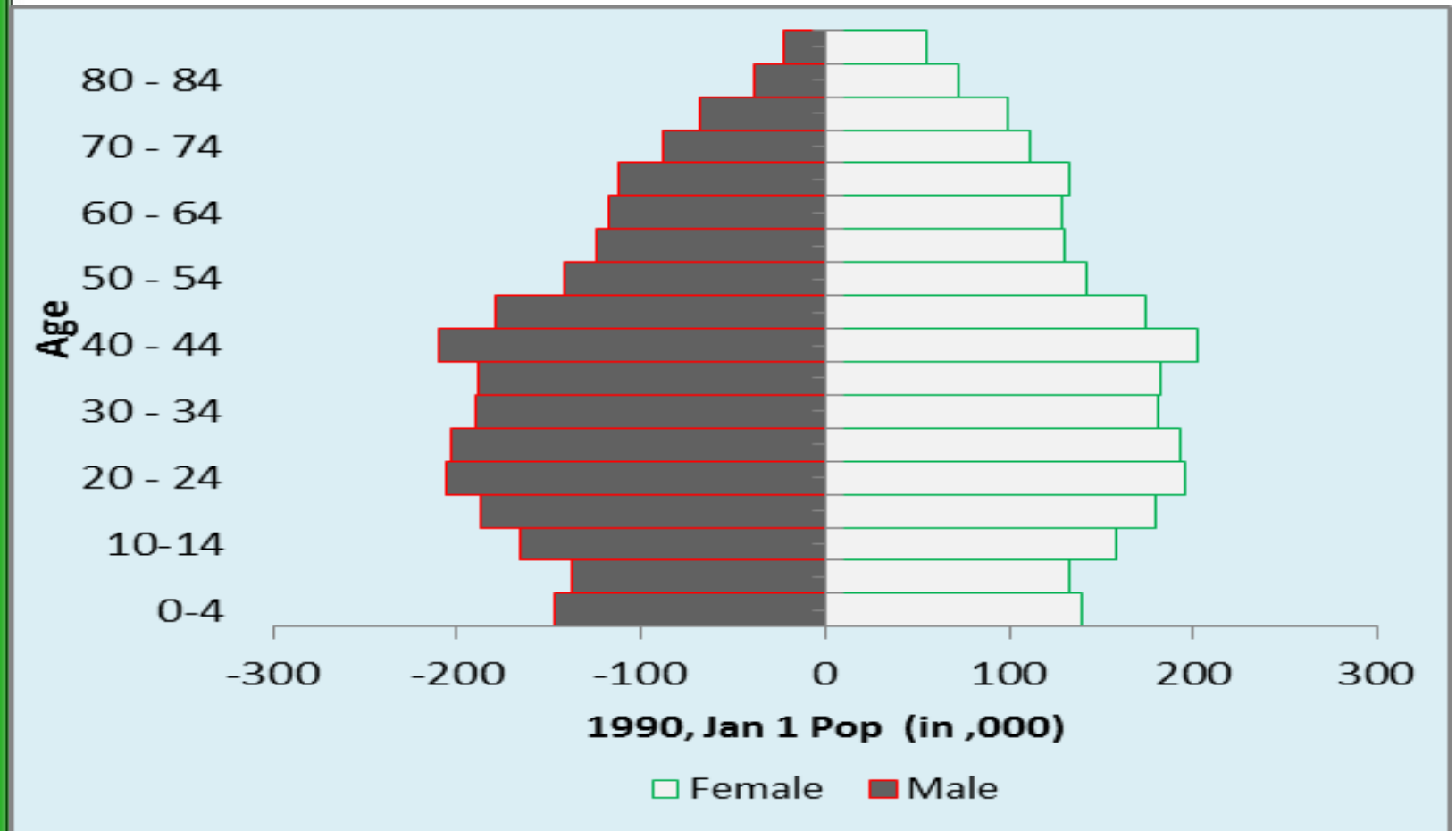
- Dependent on accurate data input from tables and figures
- Can be difficult to see actual numbers, and limited to the number of lines in a graph

Population Pyramid

- A population pyramid is a graphical illustration that shows the distribution of various age groups in a population. It is used to depict the age-sex structure of a population



Population Pyramid



http://www.ined.fr/en/everything_about_population/graphs-maps/population_graphs/



Key benefits of data presentation using tables and graphs

Any graph used to report finding should show:

- Significant features and findings of the investigation (e.g., M&E work) in an easily read way
- Relationships between and within variables
- Data profile and communicate findings to planning groups and decision making

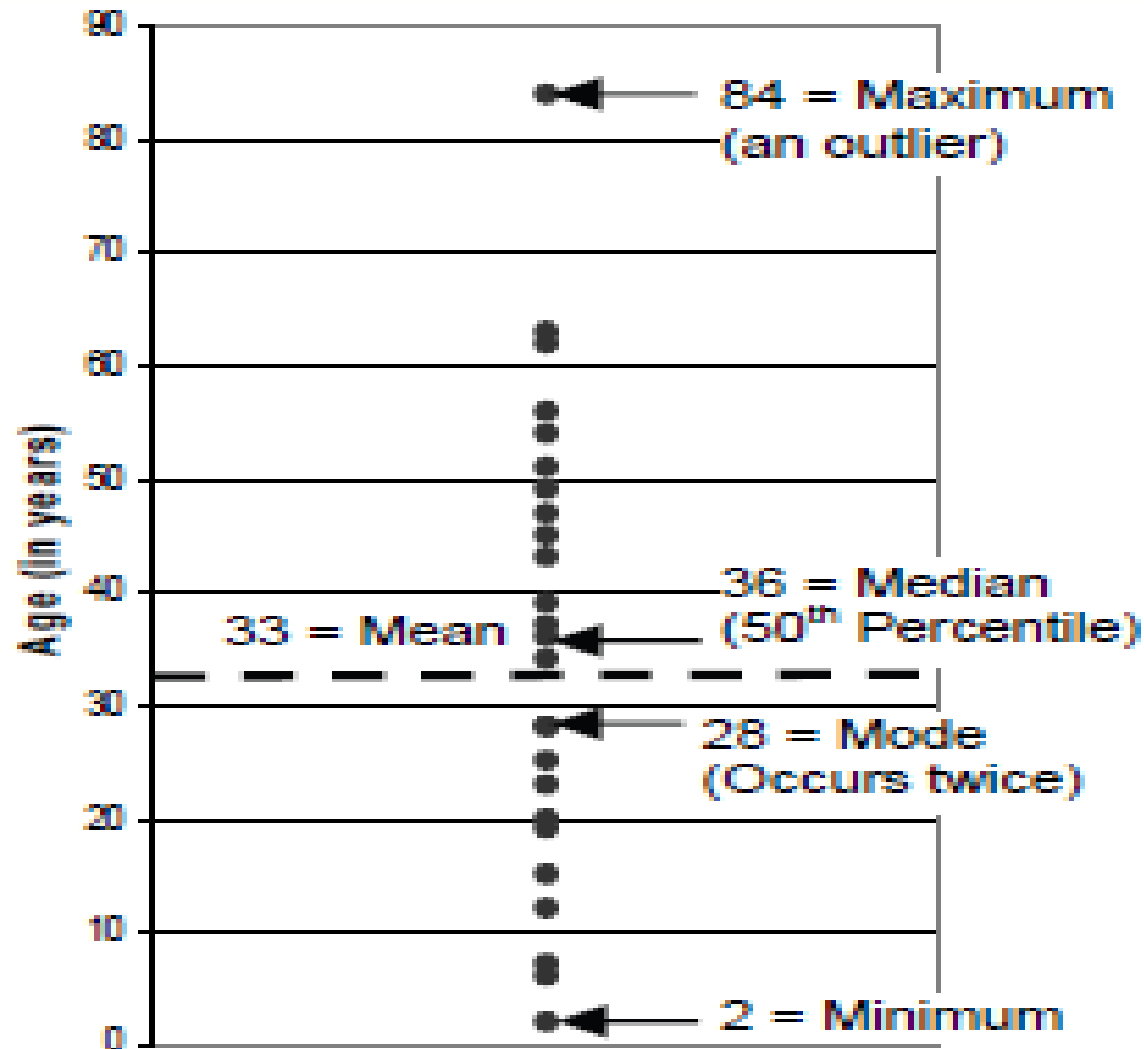


Summarizing **Continuous** Variables

- Mean
- Median
- Mode
- Range
- Standard deviation



Statistics describing continuous variable distribution



Mean

Mean = $\frac{\text{sum of value}}{\text{\# of observations}}$

Example: age (yrs) of children (8, 11, 10, 8, 9)

$$\text{Mean: } \frac{\text{sum}}{\text{\# obs.}} = \frac{8+8+9+10+11}{5} = \frac{46}{5} = 9.2 \text{ yrs}$$

Mean

Mean = $\frac{\text{sum of value}}{\text{\# of observations}}$

Example: age (yrs) of individuals (8, 56, 10, 8, 9)

$$\text{Mean: } \frac{\text{sum}}{\text{\# obs.}} = \frac{8+8+9+10+56}{5} = \frac{91}{5} = 18.2 \text{ yrs}$$

➤ Adding an outlier (age = 56) changes the mean dramatically!



Median

Median = Middle value

Example: age (yrs) of children (8, 11, 10, 8, 9)

Median: -Order data: 8, 8, 9, 10, 11

-Pick the middle value

Here it is the 3rd: 9 years

Mode

Observation that occurs most frequently

9 12 15 15 15 16 16 20 26

Observation	Number of occurrences
9	1
12	1
15	3
16	2
20	1
26	1

Range

- The spread of the data from lowest (minimum) value to highest value (maximum)

Example:

9 12 15 15 15 16 16 20 26

Range = (9—26)

Sometimes reported: Range = 17

Standard Deviation (σ)

- Measure of the deviation or distance of the observations from the mean
- Standard deviation = $\sqrt{\frac{\sum (x_i - \bar{x})^2}{N - 1}}$



Standard Deviation (σ)

Example- Calculate the standard deviation for the following set of numbers:

9 12 15 15 15 16 16 20 26

$$\text{Standard deviation} = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N - 1}}$$

$$\text{Mean} = 16$$

$$N = 9$$

$$[9-16]^2 + [12-16]^2 + [15-16]^2 + [15-16]^2 + [15-16]^2 + [16-16]^2 + [16-16]^2 + [20-16]^2 + [26-16]^2 = 184$$

$$\text{SD} = \sqrt{\frac{184}{8}} = 4.79$$



Data Sources and Utilization Level

Utilization Level	Type of Data	Sources
Inputs (Resources)	Administrative Data	Administrative Records (E.g. Hospital Records- Health Statistics)
Process (Activities)	Administrative Data	Administrative Records
Program Outputs (E.g. # of Clinics built)	Administrative Data	Administrative Records
Intermediate Outcomes (E.g. Access to Health Care)	Administrative Data & Surveys (ZDHS, ZAMPHIA, LFS)	Administrative Records (MOH), CSO
IMPACT	Surveys (ZDHS, LFS etc), Census	CSO



EVALUATION OF DEMOGRAPHIC DATA



Overview

- Data are used in various aspects that encompass all aspects of human, economic, social and political development.
- Various sources are used to capture social, economic and demographic data such as the **census, surveys, vital registration systems and population registers** among others



Overview

- In developing countries the common sources are the censuses and surveys.
- Despite surveys and census being the dependable sources of data, their **reliability, Accuracy can not be fully be guaranteed**
- Thus, social, economic and demographic data need to be evaluated before it is used.



Why Evaluate Data?

- To identify **sources** and **types of errors** in order to know which sections of data **contain errors** and what **procedures** of data collection and processing lead to these error, useful in planning for future preventive measures
- To determine the accuracy of data collected
- To **make appropriate adjustment** to the data before we can use confidently to estimate demographic events, plan for social and economic development.



Principles Sources of Errors in Data

- Errors resulting from **inadequate preparation in data collection, collation, analysis**
- Errors committed during **data collection**
- Errors due to **processing of data** (editing, coding, transcription, entry)
- Sampling errors
- Non Sampling Errors



Non Sampling Errors

- These are errors that arise during the course of all data collection activities and processing procedures
- These exist in both sample surveys and censuses data.
- Non-sampling errors can be divided into two main categories:
 - **Coverage errors**
 - **Content errors**



Types Sources of Errors in Data

A. Coverage Errors

- These are as a result of either under enumeration/over enumeration of a place, village, or persons
- They occur when there is an omission, duplication or erroneous inclusion of units in the sampling frame.

B. Content Errors

- These arise as a result of;
- Interviewer biases
- Use of wrong questionnaires
- Misunderstanding of questions, memory lapse, deliberately giving wrong answers
- Data entry
- Response, Non response, data processing and reporting

Common Errors in data

- Errors in demographic data affects the composition of the population, such as age, marital status and other characteristics.
- **Misstatement of age**
 - Underreporting (or under-enumeration) of age, measuring coverage or completeness for age enumeration
 - Misreporting of age, and
 - Non-reporting of age
- In some developed countries, misreporting of sex is negligible, for there appear to be little or no reason for a tendency for one sex to be reported at the expense of other.



Age and Sex Data

- Measurement of Age and Digit Preference
- Types of Age errors
- Graphical Analysis
- Indexes of age preference
- Whipple's Index
- Myers' blended index
- Age ratio analysis



Methods of Detecting Errors

- Two methods;

A. Direct Method

- This method entails having two independent methods that should collect the same information e.g the census and the post enumeration survey. The post enumeration survey is done on a selected area. The results of the post enumeration survey are compared with the results of the census in that selected area. The difference between the two reflects the margin of error/accuracy



Methods of Detecting Errors

B. Indirect Method

- **External consistency** check method
(Involves comparing the results from one census to another or the census with other sources e.g survey, VRS
 - Balancing equation
 - Error of closure
 - Intercensal growth rate



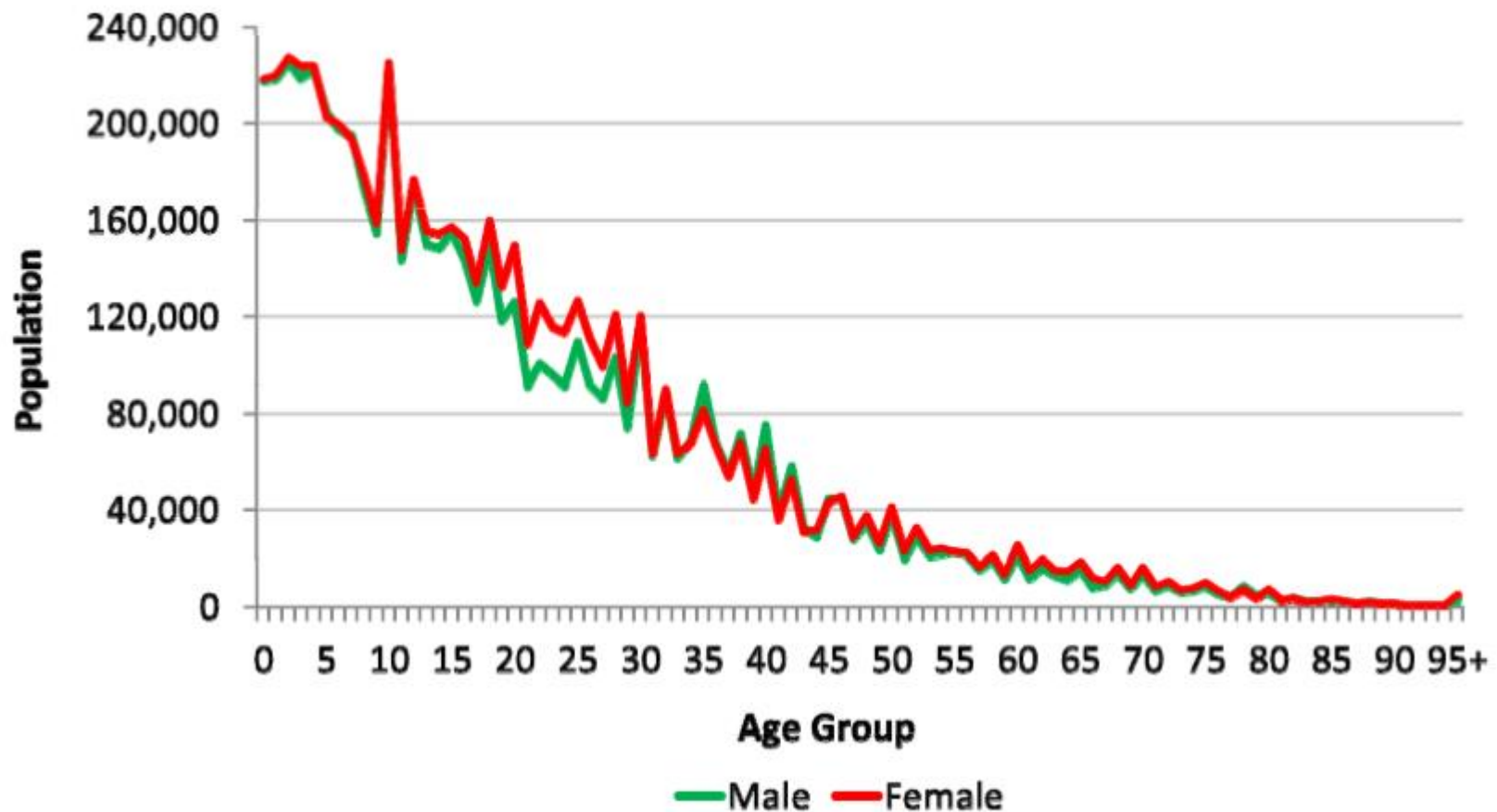
Indirect Methods Cont...

- **Internal Consistence Check;** this method rely heavily on age sex data, this is because age and sex influence a lot of social, economic and demographic data
 - **Single year age;** this method detects errors as a result of age misstatement, age preference in numbers ending with 0, 5 even numbers. Under this method the following are used;
 - Graphical method which provides a visual inspection of the data
 - Mathematical methods which involves methods of deriving indices for digit preference.



Graphical Method

Figure 13.5: Population Distribution in Single Years, Zambia 2010



Source: 2010 Census of Population and Housing

Methods used to detect digit preference (Errors in data)

Index of age preference

- **Whipples index;** This method was developed to detect the degree of age misstatement that can be attributed to the preference of age ending in digit 0 and 5
- **Myers Blended index;** this method was developed to detect the preference of all digits ending 0 to 9. This method gives equal weight in terms digit preference in all the digits.



Methods of Smoothing/Graduating data

- Graphical methods
- Mathematical methods
 - Moving averages
 - UN 5 point Method
 - Carrier-Farag method
 - Sex Ratio Method

